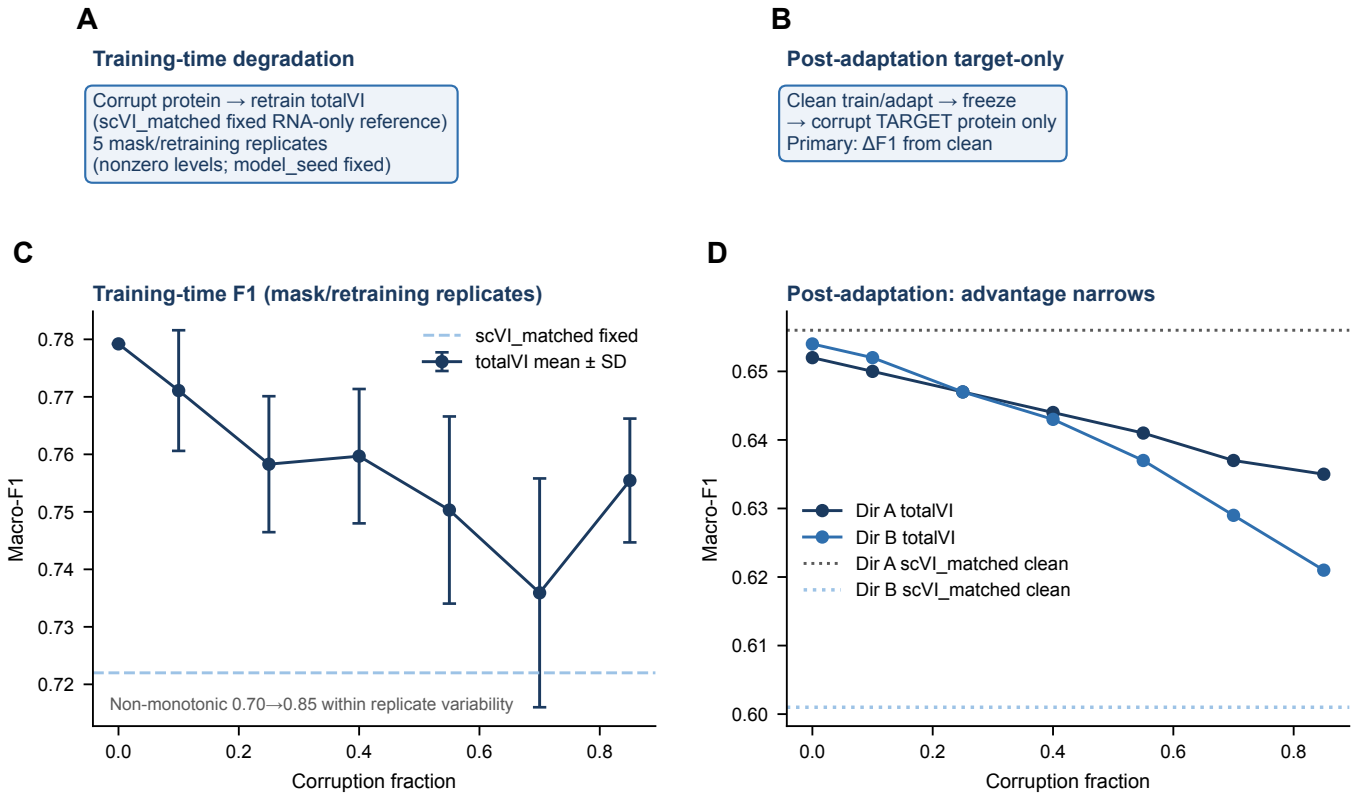
A
Training-time degradation
Corrupt protein → retrain totalVI
(scVI_matched fixed RNA-only reference)
5 mask/retraining replicates
(nonzero levels; model_seed fixed)
B
Post-adaptation target-only
Clean train/adapt → freeze
→ corrupt TARGET protein only
Primary: ΔF1 from clean
C
Training-time F1 (mask/retraining replicates)
scVI_matched fixed
totalVI mean ± SD
Macro-F1
0.78
0.77
0.76
0.75
0.74
0.73
0.72
Non-monotonic 0.70→0.85 within replicate variability
0.0
0.2
0.4
0.6
0.8
Corruption fraction
D
Post-adaptation: advantage narrows
Macro-F1
0.65
0.64
0.63
0.62
0.61
0.60
Dir A totalVI
Dir B totalVI
Dir A scVI_matched clean
Dir B scVI_matched clean
0.0
0.2
0.4
0.6
0.8
Corruption fraction